

Does How We Talk About LLMs Matter? Effects of Mentalistic and Mechanistic Language on User Perceptions

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Abstract

The rapid adoption of large language models (LLMs) has raised concerns that describing these systems using mentalistic language may encourage anthropomorphism and distort users' perceptions. In this work, we experimentally tested whether such mentalistic versus mechanistic descriptions of ChatGPT influence mental state attributions, trust, and responsibility judgments. In a preregistered between-subjects experiment ($N = 500$), participants were randomly assigned to read a mentalistic, mechanistic text, or no text (control) before reporting their perceptions of ChatGPT. Across mental state attribution, trust, and responsibility attributions, language showed limited effects. Mentalistic language increased attributions of agency but did not affect other mental state dimensions, trust, or responsibility attributions. Mechanistic language did not reduce anthropomorphic perceptions relative to the control condition. Instead, prior LLM usage consistently predicted higher mental state attributions and trust. These findings suggest that short-term linguistic framing has limited influence on perceptions of LLMs, whereas users' experience with AI plays a more central role.

Keywords

LLM, mentalist vocabulary, trust, agency, anthropomorphism

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1 Introduction & Related Work

The rapid emergence of large language models (LLMs) has renewed long-standing concerns about how intelligent technologies are described and understood. In particular, several works [2, 19, 29, 33] have cautioned against the use of mentalistic language (language to express mental states such as "think", "understand", "know" or "believe") to describe LLMs, arguing that such language may promote anthropomorphic interpretations that obscure the systems' underlying statistical mechanisms. To that end, the central concern is not merely semantic: anthropomorphism, that is, the attribution of human-like attributes, has been linked to shifts in trust,

responsibility attribution, and moral evaluation of artificial agents [9].

Anthropomorphism of technology is, however, neither new nor limited to LLMs. With the growing digitization and the ongoing technological progress, anthropomorphism of technologies has extensively been studied for various technologies [3, 18, 21, 26, 32, 37], suggesting that anthropomorphism is a robust and recurring phenomenon rather than an anomaly specific to contemporary AI. Recent work has begun to examine anthropomorphism in the context of LLMs specifically, with early evidence suggesting that their advanced language capabilities facilitate human-like perceptions [7, 8, 15, 28, 39]. Most of these works connect LLMs' language abilities to increased human-like perceptions, arguing that "language serves as arguably one of the most direct and perceptible reflections of others' conscious minds" [25, p.3]. Additionally, first studies have also found that the extent to which users anthropomorphise LLMs varies depending on certain design cues [1, 7, 27, 40] and application domains [7, 14]. For example, Cohn et al. [7] found that when LLMs responded with both text and speech, anthropomorphism increased, resulting in increased perceptions of information accuracy and decreased perceptions of risk.

Moreover, AI in general but LLMs in specific are built to mimic human behavior (here: speech). Hence, anthropomorphising LLMs might be less surprising, considering that these systems are built to replicate human speech. To that end, LLMs outperform previous rule-based chat systems in their ability to produce language, so far that the produced text cannot be differentiated from human-written text, including persuasive [5], empathetic [35], and deceptive text [23]. This inherent anthropomorphic abilities might as well cancel out any calls to caution against anthropomorphism as any interaction "looks and feels every bit like interacting with another human" [24, p. 5].

Hence, while the anthropomorphism of LLMs might be less surprising, it is less clear whether this is good or bad. In their work, Peter et al. [24] conclude that anthropomorphism is a double-edged sword including good and bad consequences, such as making information-dense text more accessible [31], supporting tutoring through personalized conversation style [22], but also allowing for manipulation and deception at scale [23].

Overall, this raises an important question: **If anthropomorphism of technologies in general and LLMs in specific is so common that it might be unavoidable to do so, why would the way we talk about LLMs matter?**

Researchers from various fields such as linguistics, psychology, philosophy, and anthropology, have long studied the relationship between language production and the way we think, suggesting



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that the structure of a language influences its speakers' worldview and cognition [4, 6, 16, 36].

For the context of technology, empirical studies could show that, indeed, how we talk about technologies affects how users perceive them. For example, calling something an artificial intelligence, a machine learning algorithm, or a computer program has been shown to affect laypeople's perceptions (e.g., perceived complexity) and evaluation (e.g., trust) of system properties [17]. Likewise, Epstein et al. [10] could demonstrate that changes in language (i.e., "tool" or "agent") could alter the way that people think about these systems by increasing perceptions of anthropomorphism and attributions of responsibility. Resonating with these results, Waytz et al. [34] found that when autonomous vehicles were described in more anthropomorphic language, participants trusted them more and allocated more responsibility to the car than to the driver.

For this work, we follow the above-reviewed body of work, and suggest that the way we talk about LLMs matters. Concretely, we hypothesize that:

H1: Participants who read a text that uses *mentalistic* vocabulary to describe ChatGPT will be *more* likely to (a) attribute mental states to ChatGPT, and (b) trust ChatGPT, than those who do not read such a text.

In addition, we also wanted to examine if more mechanistic language would drive perceptions in the opposite direction. Concretely, we hypothesized that:

H2: Participants who read a text that uses *mechanistic* vocabulary to describe ChatGPT will be *less* likely to (a) attribute mental states to ChatGPT, and (b) trust ChatGPT, than those who do not read such a text.

Going beyond attributions of psychological capacities and anticipated reliability, we were also interested whether language would affect how more normative judgments. Hence, in an open research question we additionally explored: RQ: What are the effects of mentalistic and mechanistic descriptions on the attribution of responsibility to the system (ChatGPT), the programmers, and the seller (OpenAI)?

2 Method

This study received ethical approval of the ethics board of the University of [blinded for review] under number 2506SPMW0444. All hypotheses were preregistered and can be found here¹. All data, analyses, and code are publicly available here.

2.1 Experimental design and procedure

To test our hypotheses, we conducted a three-group between-subjects design. $N = 500$ participants were recruited through the online panel provider Prolific, and were randomly assigned to one of the three conditions: control, mentalistic, and mechanistic. To gain the baseline mental attributions to and trust in ChatGPT, participants in the control condition directly answered all measures. Participants in the two experimental conditions either read a text using mentalistic vocabulary (214 words) or mechanistic vocabulary (223 words). To

ensure that both texts were as comparable as possible, only single words and phrases were exchanged. The text are provided in Supplementary Material A.

After being briefed about the aim of the study and providing informed consent, participants in the two experimental conditions were asked to read a short text, describing how the LLM ChatGPT functions (see Supplementary Material A). Immediately after reading, we asked participants to indicate the fluency of the text by asking on a seven-point Likert scale "To what extent was this text easy or difficult to understand?". Results of a t-test indicated that the mechanistic text was more difficult for participants to read than the mentalistic text ($t(333) = 7.50, p < .001$ ($M_{\text{mental}} = 5.90, M_{\text{mechanic}} = 4.87$). Hence, fluency ratings were included as confounders in later analyses. After reading the respective text, participants provided their answers for trust, responsibility attribution, and mental state attribution. The study closed with a debriefing.

2.2 Measures

2.2.1 Mental state attribution. To assess participants' mental state attribution, we employed two questionnaires. To that end, the scale by Gray et al. [12] captures mental state attribution via the two dimensions agency (seven items, e.g., "To what degree is ChatGPT capable of making plans and working toward goals?") and experience (eleven items, e.g., "To what degree is ChatGPT capable of feeling afraid or fearful?"). Answers were collected on a five-point Likert scale ranging from "not at all" to "completely". Reliability for both subscales was good (agency: Cronbach's $\alpha = .82$; experience: (Cronbach's $\alpha = .96$).

Like the previous scale, the scale by Miraglia et al. [20] differentiates two dimensions: Mental states with neutral or positive valence (e.g., "Can a system learn?"), and mental states with negative valence (e.g., "Can a system tell a lie?"). Answers were collected on a five-point Likert scale ranging from "not at all" to "completely". Reliability for both subscales was acceptable and good, respectively (neutral or positive: Cronbach's $\alpha = .75$; experience: (Cronbach's $\alpha = .87$).

2.2.2 Responsibility attribution. To assess responsibility attributions, we followed Tolmeijer et al. [30] and asked participants the following questions: "To what extent do you hold [entity] responsible for collateral damage?". As entities, we asked participants to about their responsibility attributions to ChatGPT, the programmers' of ChatGPT, and OpenAI as the organization behind ChatGPT. Answers were collected on a five-point Likert scale ranging from "not at all" to "completely".

2.2.3 Trust. To assess trust in ChatGPT, we used the global trust dimension of the Trust in AI Scale (TAIS) by [38] that consists of the five items (e.g., "I believe in ChatGPT"). Answers were collected on a five-point Likert scale ranging from "strongly disagree" to "strongly agree". Reliability of the scale was good (Cronbach's $\alpha = .90$).

2.3 Sample

The sample comprised 507 UK citizens (232 female, 269 male, 2 non-binary, 4 undisclosed), all at least 18 years old (M age = 41.73, $SD = 13.3$, range 18–79). Regarding education, 91 participants reported

¹Additional hypotheses are reported in OSF but not reported here due to space limitations. Results of the additional hypotheses can be found in OSF

Table 1: Descriptive statistics of trust, mental state attributions, and perceived understanding (divided into perceived understanding and perceived transparency).

	control		mentalistic		mechanistic	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
trust	3.43	0.80	3.40	0.92	3.28	0.84
agency	2.96	0.80	3.14	0.83	2.95	0.83
experience	1.31	0.52	1.32	0.54	1.30	0.47
AMS_NP	2.29	0.59	2.37	0.65	2.34	0.63
AMS_N	2.13	0.68	2.06	0.72	2.22	0.66

Note. AMS_{NP} = Mental states with neutral or positive valence; AMS_N = Mental states with negative valence.

no professional qualification, 134 some vocational training, and 294 held a university degree, with the remainder having completed other forms of professional training (e.g., industrial, agricultural, commercial, or internships).

Additionally to these demographic data, we also asked participants if they had used a LLM before. Of the 500 participants who answered the LLM-use question, 469 had used an LLM before and 31 had not. Among prior users, 106 reported using LLMs more than once a day, 115 once a day, 125 once a week, 39 every other week, and 77 once a month.

3 Results

Analysis was performed using R version 4.5.1. All analyses were preregistered and can, together with all data, be found here. We provide an overview of the descriptive results in Table 1.

3.1 Hypotheses testing

3.1.1 Effects of language on mental state attributions. In H1 and H2, we predicted that reading a text using either mentalistic or mechanistic vocabulary would induce more or less mental state attribution than in the no text group (control), respectively. To test this, we ran four separate multiple linear regressions for each of the four measures of mental state attribution. For the experimental grouping variable as the predictor, the control group was always the reference group. Additionally, we included in all four regressions the fluency scores and participants' usage scores.

For agency, the overall model was significant, $F(8,491) = 7.09$, $p < .001$, explaining 10.4% of the variance in agency ($R^2 = .104$, adjusted $R^2 = .089$). The experimental factor was a significant predictor, with participants in the mentalizing group reporting higher agency than the control group ($b = 0.64$, $SE = 0.22$, $t = 2.91$, $p = .004$), supporting H1. However, unlike hypothesized, participants in the mechanistic group reported similar levels of agency as the control group ($b = 0.34$, $SE = 0.19$, $t = 1.83$, $p = .068$). Additionally, we found that the control variable, participants' LLM usage frequency, also predicted agency attributions: higher usage levels were associated with greater agency perception ($b = 0.34$ – 0.57 , $p < .017$).

In contrast to these results, for the second dimension, experience, the overall model was not significant, $F(8, 491) = 0.64$, $p = .75$, $R^2 = .01$, adjusted $R^2 = -.01$. The experimental conditions had no effect on mental state attributions (mentalistic: $b = 0.02$, $SE = 0.14$, $p = .88$; mechanistic: $b = -0.01$, $SE = 0.12$, $p = .94$).

Similarly, the experimental conditions did not affect the two dimensions of the mental state attribution scale by Miraglia et al. [20]. For the AMS_N dimension, the model was not significant, $F(8,491) = 0.86$, $p = .548$, ($R^2 = .014$, adjusted $R^2 = -.002$). While for the AMS_{NP} dimension, the model was significant, $F(8,491) = 3.04$, $p = .002$, ($R^2 = .047$, adjusted $R^2 = .032$, this was driven by the control variable LLM usage. Like reported above, higher usage levels were associated with greater perception of mental states with neutral or positive valence ($b = 0.34$ – 0.41 , $p < .002$).

3.1.2 Effects of language on trust perceptions. In addition to mental state attributions, we predicted that reading a text using either mentalistic or mechanistic vocabulary would induce more or less trust than in the control group, respectively. To test this, we ran a multiple linear regression with trust as the dependent variable, and our experimental conditions as the predictor with the control group as the reference group. Similar to above, we included fluency and participants' usage frequency in the regression. The overall model was significant with $F(8,491) = 19.84$, $p < .001$, explaining 24.4% of the variance in trust ($R^2 = .24$, adjusted $R^2 = .23$). Unlike hypothesized, this was not due to our experimental manipulation (mentalistic: ($b = -0.20$, $SE = 0.21$, $p = .35$; mechanistic: $b = -0.32$, $SE = 0.18$, $p = .07$). Instead, we found that trust perceptions were mostly associated with usage pattern: more frequent AI usage was strongly associated with higher trust ($b = 0.41$ – 1.35 , $p < .006$).

3.1.3 Effects of language on responsibility attributions. Because responsibility attributions were measured via a single item, we conducted ordinal logistic regressions to assess whether reading a text using either mentalistic or mechanistic vocabulary would induce difference in responsibility attributions. In all three ordinal logistic regressions, the dependent variable was responsibility attribution to either ChatGPT, its developers, or its owner, modeled with a proportional odds (logistic) link function. The control group was always the reference group. Statistical significance was assessed using Wald z statistics and likelihood ratio tests (threshold for statistical significance: $|z| < 1.96$).

For responsibility attributions to ChatGPT, our results indicated no significant differences. Specifically, the effect of mentalistic language was very small and not significant ($b = -0.02$, $SE = 0.20$, $t = -0.08$). Similarly, mechanistic language had no effect ($b = -0.27$, $SE = 0.20$, $t = -1.35$). The overall model showed adequate fit (Residual Deviance = 1391.08, AIC = 1403.08).

Results for responsibility attributions to developers and OpenAI mirror these results. For the developers, mentalistic language was associated with higher responsibility attributions which was, however, not statistically significant ($b = 0.26$, $SE = 0.20$, $t = 1.32$). The effect mechanistic language was non-significant ($b = -0.01$, $SE = 0.19$, $t = -0.04$). The overall model fit was acceptable (Residual Deviance = 1543.50, AIC = 1555.50). For OpenAI, mentalistic language was associated with higher responsibility attributions but did not reach statistical significance ($b = 0.26$, $SE = 0.20$, $t = 1.33$), suggesting a tendency toward higher responsibility attributions compared to the control group. The coefficient for mechanistic language was small and likewise non-significant ($b = 0.03$, $SE = 0.19$, $t = 0.17$).

4 Discussion

In this paper, we tested the effects of mentalistic and mechanistic language to describe LLMs (here: ChatGPT) on (a) users attribution of mental states to these models, (b) trust, and (c) attributions of responsibility. For mental state attributions, the experimental manipulation affected only one of four dimensions. Participants exposed to mentalistic language attributed higher agency to ChatGPT than those in the control group. However, we observed no effects of language for the experience dimension or for the two dimensions of the mental state attribution scale by Miraglia et al. [20]. Similarly, participants' trust perceptions and responsibility attribution were not affected by the language manipulation, supporting previous null findings [14]. A consistent finding across mental state and trust measures was the role of prior AI usage. Participants who reported more frequent use of large language models were more likely to attribute mental states to ChatGPT and indicated higher trust compared to those who use ChatGPT less.

We draw several conclusions from these results. First, although our interventions was arguably rather weak (one time exposure), we could still see an effect for the agency dimension. We see this as evidence that language indeed, as found in other cases, alters perceptions. Because the other dimensions were unaffected, we further assume that this perceptibility is, however, limited to specific aspects of mental state attributions, and does not include perceptions of experience such as emotional states (see Limitations section for possible exceptions), resonating previous results [11, 13]. Our results also indicate that altered mental state attributions do not translate into other attributions such as trust and responsibility, momentarily easing proclaimed worries about the risks of anthropomorphizing. To that end, our results remain inconclusive. The effects might not exist or were hidden due to the limited effect of the manipulation.

Lastly, the strong effects of familiarity and experience raise further questions. Because our design allows only for relational interpretation, we cannot comment on any direction of an effect: Does increased usage lead to higher trust, or trust to higher usage? Both seem plausible. More interesting may be results for mental state attributions. Did the frequent usage lead to greater mental state attributions, or did greater mental state attributions lead to more frequent usage? Answering this question is not trivial, if the normative goal is to reduce such arguably inappropriate attributions.

4.1 Limitations and Future Directions

As with all studies, our results come with specific limitations. First, because we only examined short-term effects of language exposure our manipulation might have not been strong enough to tease out differences between conditions. Designs with repeated exposure, possibly also using different modalities such as audio-visual content, would increase the power of the manipulation.

Second, although we relied on widely used and well-cited measures of mental state attribution, these instruments have conceptual limitations when applied to artificial systems. Several items of these scales rely on mentalistic phrasing (e.g., "the system can learn"), which may inadvertently prime anthropomorphic interpretations

and blur the distinction between participants' spontaneous attributions and the language embedded in the measurement instrument. Future research would benefit from developing measures that more carefully disentangle descriptive system capabilities from explicitly mentalistic interpretations.

Third, the content and context of the texts used in the manipulation may have constrained the observed effects. The texts focused on explaining how LLMs work, which may have implicitly encouraged a more technical or functional understanding of the system, even in the mentalistic condition. Language effects might differ in contexts where AI systems are framed in more explicitly human-like roles (e.g., as companions or social partners) or, conversely, in highly instrumental settings (e.g., financial trading or tax preparation). Thus, the generalizability of the present findings may be limited to informational contexts, and future work should examine whether stronger framing effects emerge in more socially or emotionally evocative domains.

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